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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :
(1) All Questions are *compulsory.*
(2) Answer each next main Question on a new page.
(3) Illustrate your answers with neat sketches wherever necessary.
(4) Figures to the right indicate full marks.
(5) Assume suitable data, if necessary.
(6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : **10**

(a) Define Fibre.

(b) State the meaning of $C_6H_{10}O_5$.

(c) State the source of man-made fibres.

(d) Define Spun Yarn.

(e) Name the types of twist.

(f) Mention the types of natural fibres.

(g) Draw the microscopic view of any one synthetic fibre and specify the name of the fibre.

2. Attempt any THREE of the following : **12**

(a) Differentiate between Woolen and Worsted fibres and their properties.

(b) Compare the properties of natural and man-made fibres in context to following points :

(i) Comfort

(ii) Strength

(iii) Absorbancy

(iv) Thermoplasticity

(c) Discuss the effect of twist on fabric properties.

(d) Explain Sericulture process of silk yarn manufacturing.

3. Attempt any THREE of the following : (Give reasons.) **12**

(a) Textile Industry is very important industry in India.

(b) All fibres are not considered as textile fibres.

(c) Cotton fibres are more suitable for summer season.

(d) Novelty yarns are used for fancy fabric.

4. Attempt any THREE of the following : **12**

(a) Explain the terms and the difference for the following : (Any **Two**)

(i) Amorphous and Crystalline

(ii) Thermoplastic and Non-thermoplastic

(iii) Staple and Filament

(b) (i) Mention the sources and origin of natural fibres.

(ii) Explain any one method for growing it.

(c) Explain various types of yarn : (Any **Two**)

(i) Ply yarn and cord yarn

(ii) Mono filament and Multi filament

(iii) Combed and Carded Yarns

(d) Explain the burning test of Cotton and Nylon.

(e) Explain the Retting process.

5. Attempt any TWO of the following : **12**

(a) Compare Cotton and Silk fibres in terms of its properties.

(b) Describe the manufacturing process of Viscose rayon.

(c) Explain various identification methods of fibres.

6. Attempt any TWO of the following : **12**

(a) Explain the Primary and Secondary properties of Wool and Silk fibres.

(b) Explain in detail, yarn formation process of cotton fibres.

(c) Explain with neat diagram :

(i) Types of twist

(ii) Amount of twist